SINGAPORE UNIVERSITY OF TECHNOLOGY AND DESIGN

Science, Mathematics and Technology

SSILP Math

Beginner

TRANSFORMATIONS OF FUNCTIONS, COMBINING FUNCTIONS AND INVERSE FUNCTIONS

Learning objectives

Understand Transformations of Functions

Identify and describe vertical and horizontal shifts of functions.

Combine Functions

Perform algebraic operations (addition, subtraction, multiplication, and division) on functions.

Understand and apply the concept of composition of functions to create new functions

Inverse Functions

Define inverse functions, their properties and graphical representation.

Find the inverse of a given function and verify it using the inverse function property.

Videos

[Transformations of functions (Khan academy)]
[Intro to Inverse functions (Khan academy)]

Session outline

1. Transformations of functions

2. Combining functions

What are the different ways to combine functions to make new functions?

3. Inverse functions

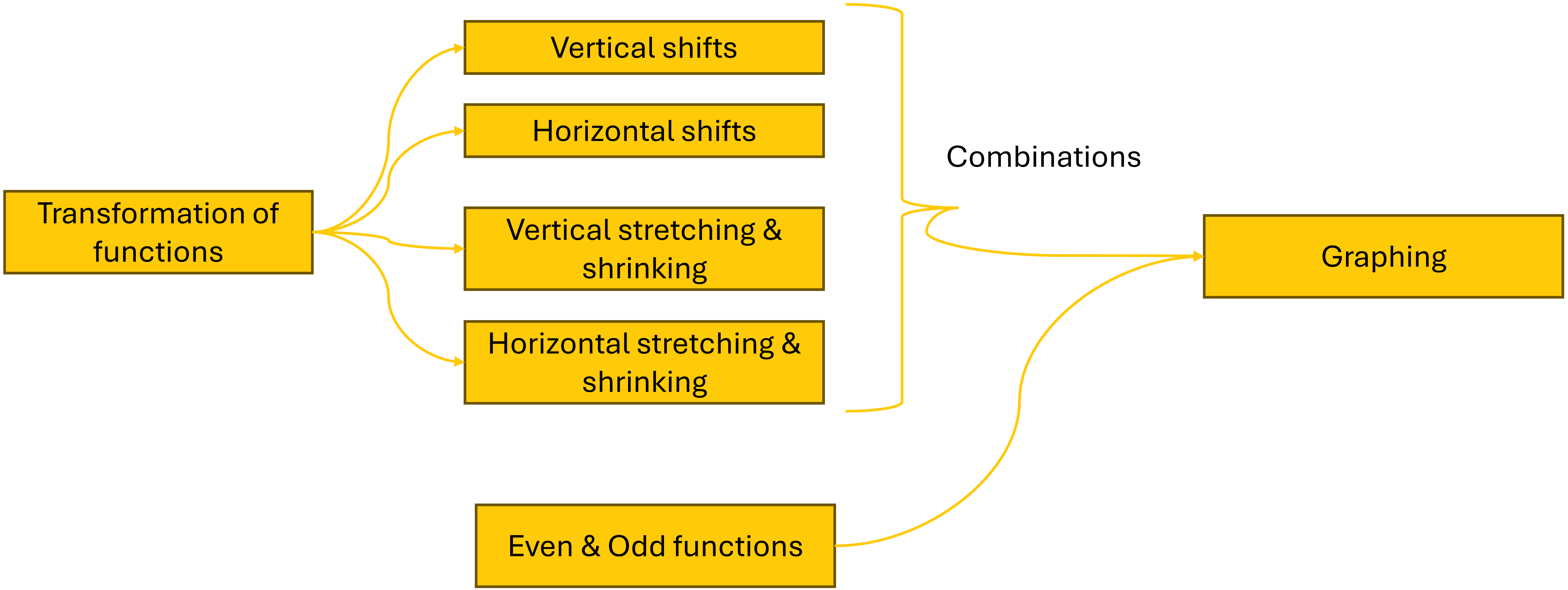
Refer to the textbook!

Pre-calculus : Mathematics for Calculus
By Stewart, Redlin, Watson (7th Ed)

Chapter 2, Section 2.6, 2.7, 2.8

1. Transformations of functions

Transformation of functions: Overview



Vertical shifting

Adding a constant to a function shifts its graph vertically:

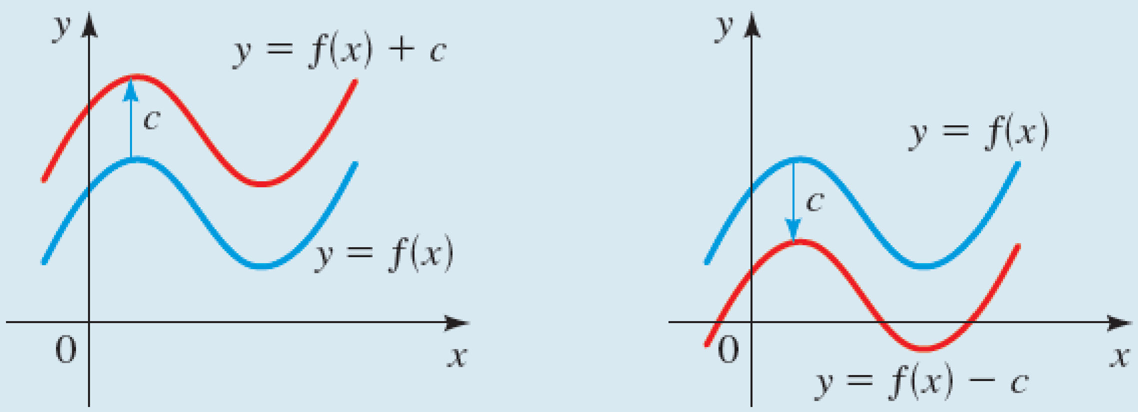
- **upward** if the constant is positive and
- **downward** if it is negative.

VERTICAL SHIFTS OF GRAPHS

Suppose $c > 0$.

To graph $y = f(x) + c$, shift the graph of $y = f(x)$ upward c units.

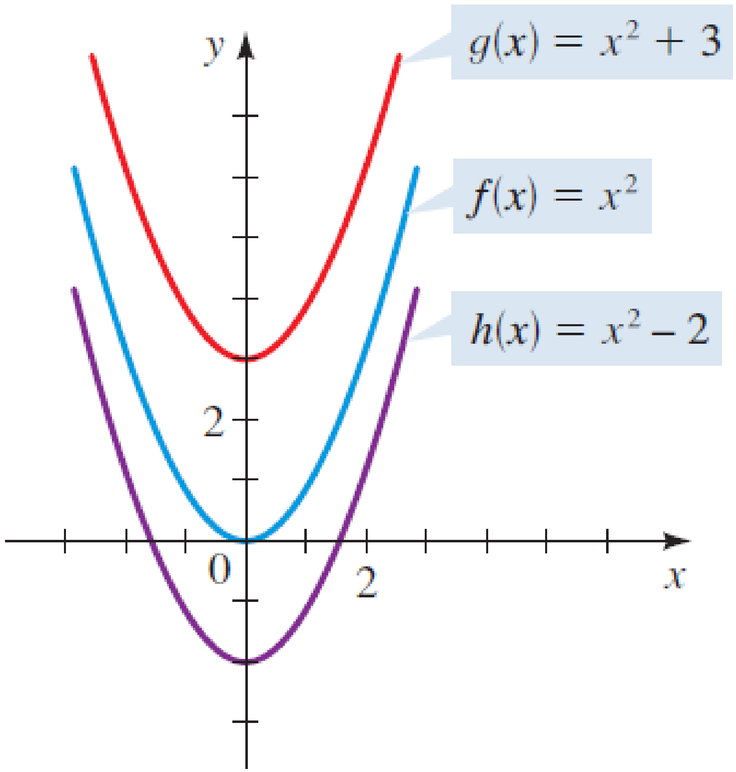
To graph $y = f(x) - c$, shift the graph of $y = f(x)$ downward c units.



Vertical shifting

Use the graph of $f(x) = x^2$ to sketch the graph of each function.

(a) $g(x) = x^2 + 3$ **(b)** $h(x) = x^2 - 2$



$$g(x) = x^2 \underline{+ 3} = f(x) \underline{+ 3}$$

y-coordinate of each point on the graph of g is **3 units above** **the corresponding point** on the graph of f.

This means that to graph g, we **shift** the graph of f **upward 3 units**

EXAMPLE

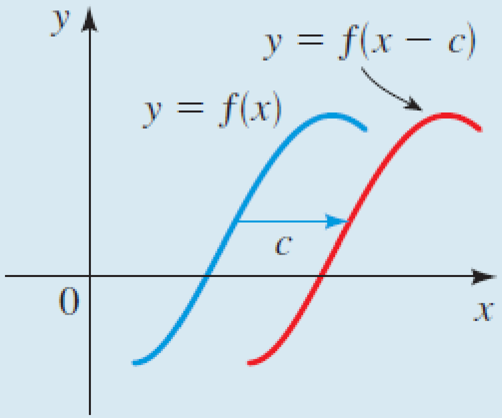
Horizontal shifting

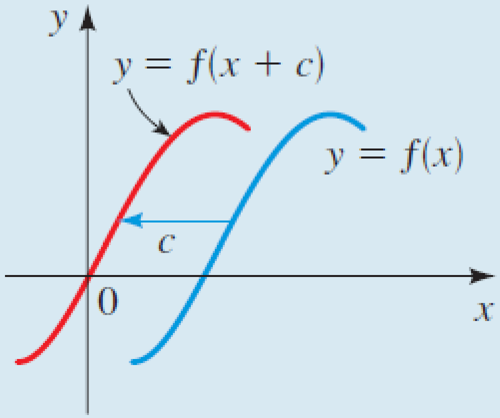
HORIZONTAL SHIFTS OF GRAPHS

Suppose $c > 0$.

To graph $y = f(x - c)$, shift the graph of $y = f(x)$ to the right c units.

To graph $y = f(x + c)$, shift the graph of $y = f(x)$ to the left c units.





Note: horizontal shift towards **right**, when we **reduce x by c** units

And vice versa

Horizontal shifting

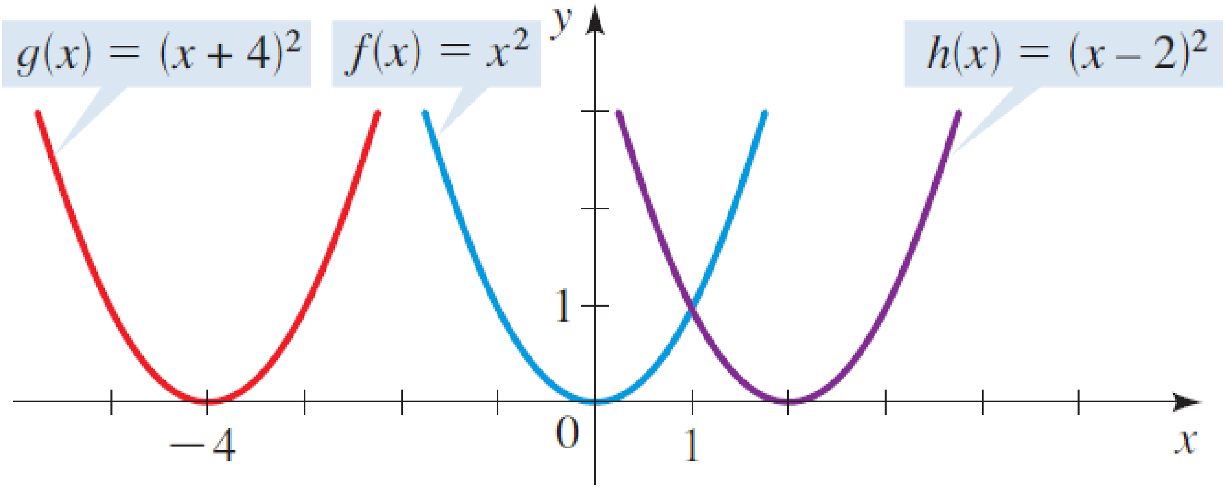
Use the graph of $f(x) = x^2$ to sketch the graph of each function.

(a) $g(x) = (x + 4)^2$ **(b)** $h(x) = (x - 2)^2$

SOLUTION

(a) To graph g, we shift the graph of f to the left 4 units.

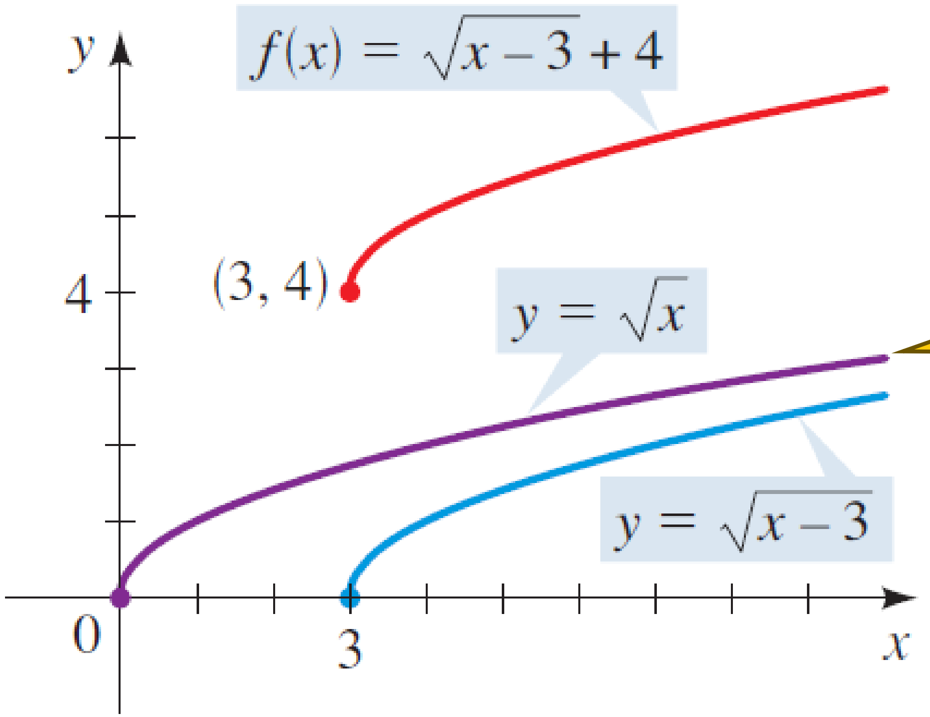
(b) To graph h, we shift the graph of f to the right 2 units.



EXAMPLE

Combining Horizontal and Vertical Shifts

Sketch the graph of $f(x) = \sqrt{x - 3} + 4$.



Start with this graph.

Shift it to the right by three units
(HORIZONTAL SHIFT)

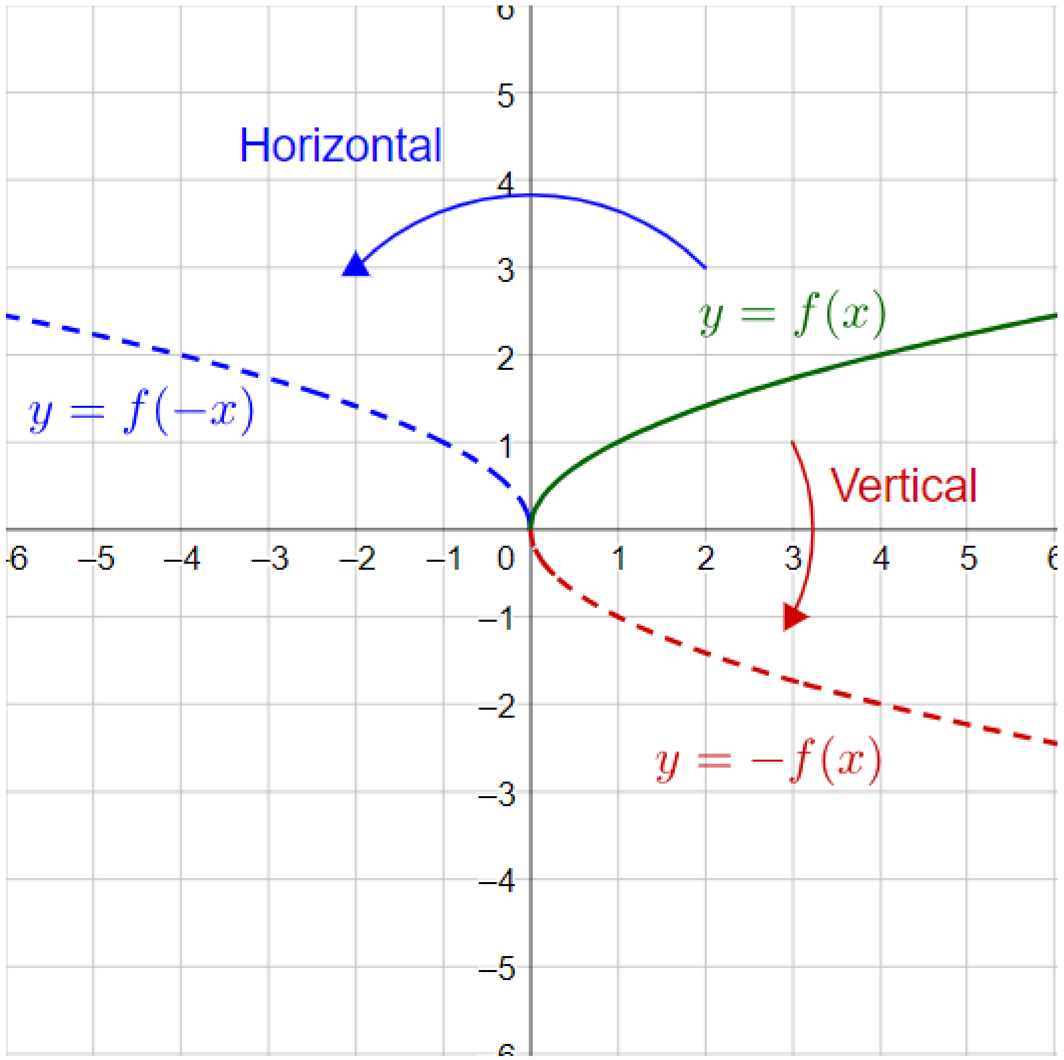
Shift resulting graph **upward by 4 units**
(VERTICAL SHIFT)

EXAMPLE

Reflecting

A **horizontal reflection** reflects the graph over the *y*-axis

A **vertical reflection** reflects the graph over the *x*-axis



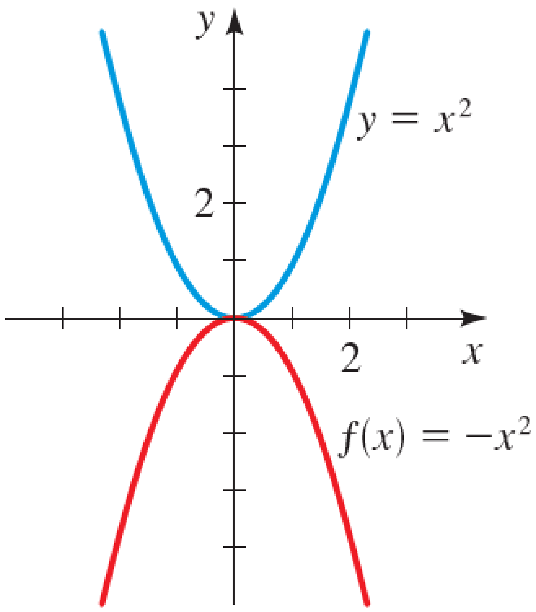
Reflecting graphs: Example

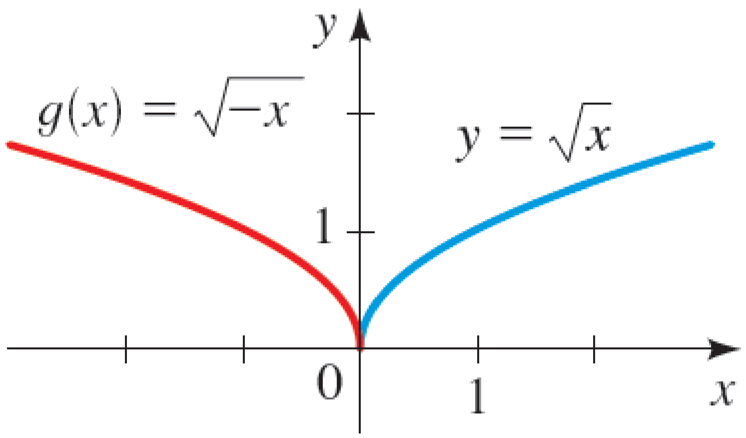
Sketch the graph of each function.

(a) $f(x) = -x^2$ **(b)** $g(x) = \sqrt{-x}$

The graph of $f(x) = x^2$ is the graph of $y = x^2$ reflected in the x -axis.

The graph of $g(x) = \sqrt{-x}$ is the graph of $y = \sqrt{x}$ reflected in the y-axis.





EXAMPLE

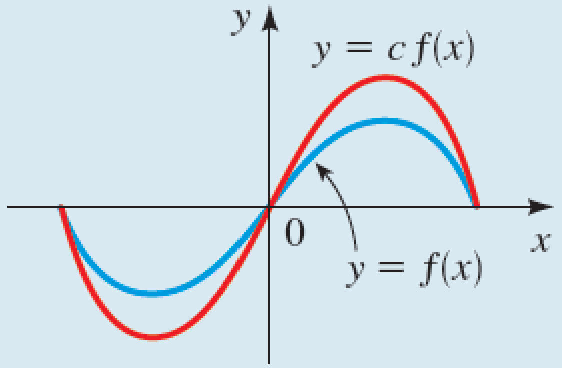
Vertical stretching and shrinking

VERTICAL STRETCHING AND SHRINKING OF GRAPHS

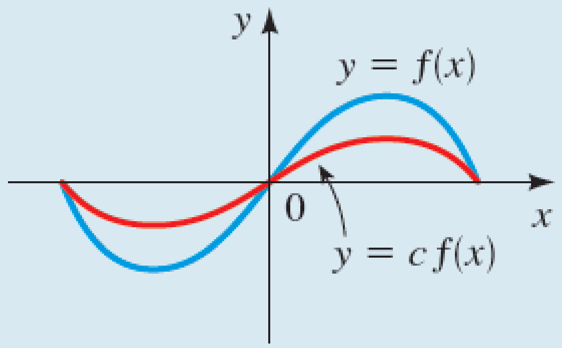
To graph $y = cf(x)$:

If $c > 1$, stretch the graph of $y = f(x)$ vertically by a factor of c.

If $0 < c < 1$, shrink the graph of $y = f(x)$ vertically by a factor of c.



$c > 1$

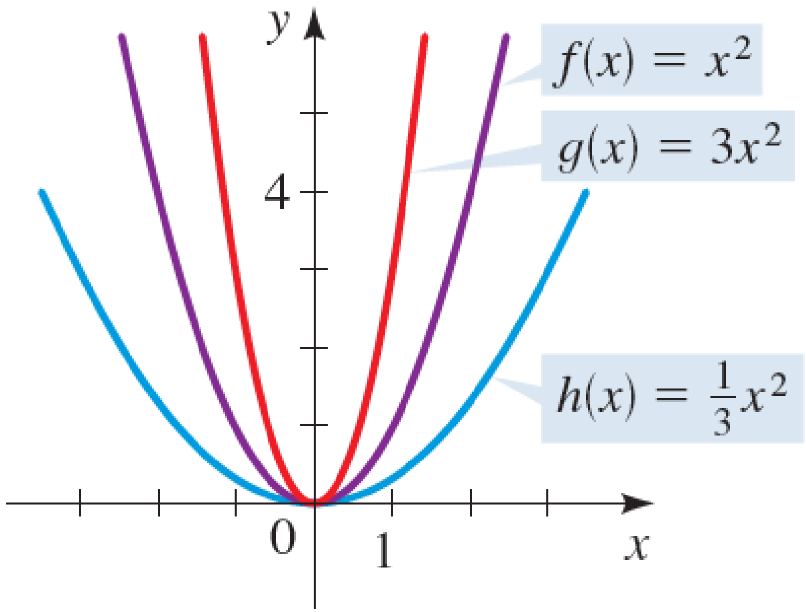


$0 < c < 1$

Vertical Stretching and Shrinking of Graphs

Use the graph of $f(x) = x^2$ to sketch the graph of each function.

(a) $g(x) = 3x^2$ **(b)** $h(x) = \frac{1}{3}x^2$



EXAMPLE

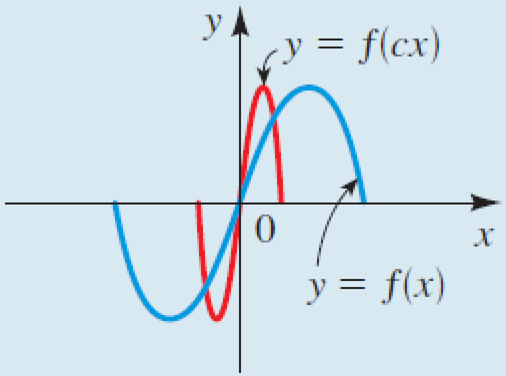
Horizontal shrinking and stretching

HORIZONTAL SHRINKING AND STRETCHING OF GRAPHS

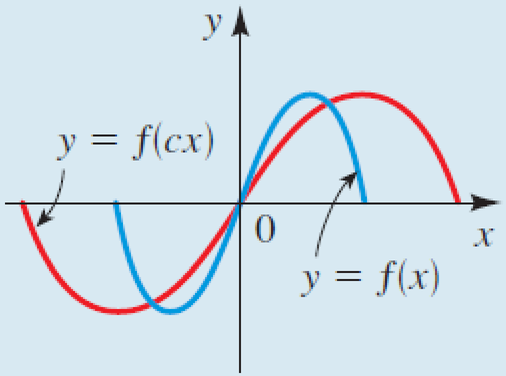
To graph $y = f(cx)$:

If $c > 1$, shrink the graph of $y = f(x)$ horizontally by a factor of $1/c$.

If $0 < c < 1$, stretch the graph of $y = f(x)$ horizontally by a factor of $1/c$.

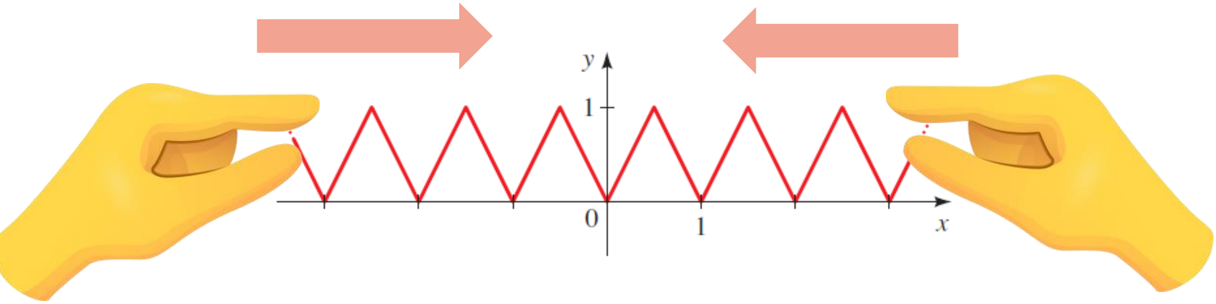


$c > 1$

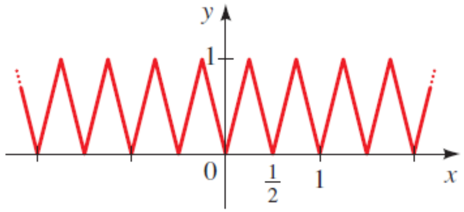


$0 < c < 1$

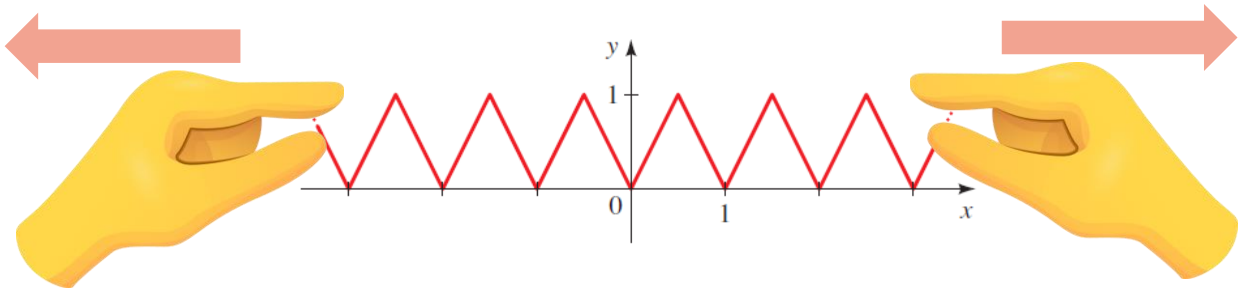
Horizontal shrinking and stretching



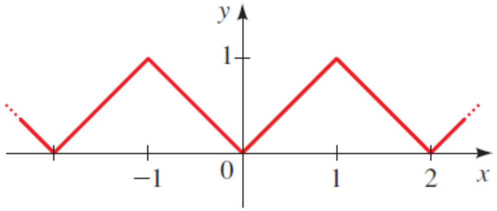
$$y = f(x)$$



$$y = f(2x)$$



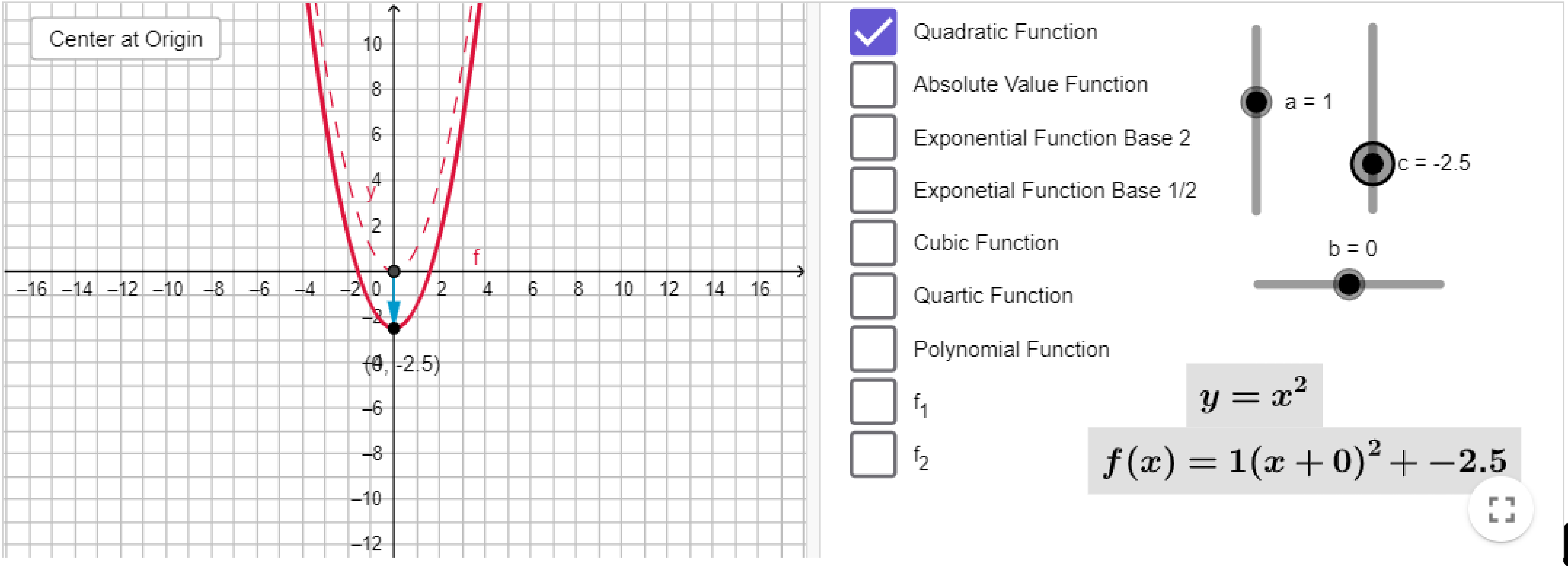
$$y = f(x)$$



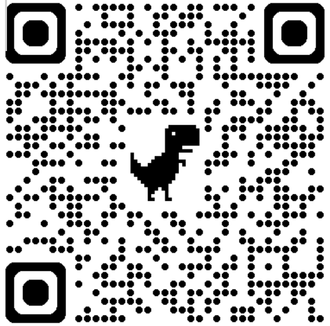
$$y = f\left(\tfrac{1}{2}x\right)$$

Transformations: Geogebra app

explore four types of function translations: reflections across the x-axis, vertical stretches, horizontal shifts, and vertical shifts



https://www.geogebra.org/m/SmqpvybK



Describing Transformations

Suppose the graph of *f* is given. Describe how the graph of each function can be obtained from the graph of *f*.

(a) $f(x - 2)$

(b) $3f(x)$

(c) $\frac{1}{3}f(x)$

(d) $f(x + 1) - 1$

ACTIVITY 1.1

1.1 Describing Transformations

SOLUTION

Graphing Transformations

Use the graph of $y = x^2$ to graph the following

(a) $g(x) = x^2 + 1$

(b) $g(x) = (x - 1)^2$

(c) $g(x) = -x^2$

(d) $g(x) = (x - 1)^2 + 3$

ACTIVITY 1.2

SOLUTION

Graphing Transformations (Other functions)

(a) $f(x) = \sqrt{x} + 1$

(b) $f(x) = -|x|$

(c) $y = \sqrt[3]{-x}$

ACTIVITY 1.3

1.3 Graphing Transformations (Other functions)

SOLUTION

1.3 Graphing Transformations (Other functions)

SOLUTION

1.3 Graphing Transformations (Other functions)

SOLUTION

Finding equations for transformations

(a) $f(x) = x^3$; shift upward 5 units

(b) $f(x) = \sqrt[3]{x}$; shift 1 unit to the right

(c) $f(x) = x^2$; shift 2 units to the left and reflect in the x-axis

ACTIVITY 1.4

1.4 Finding equations for transformations

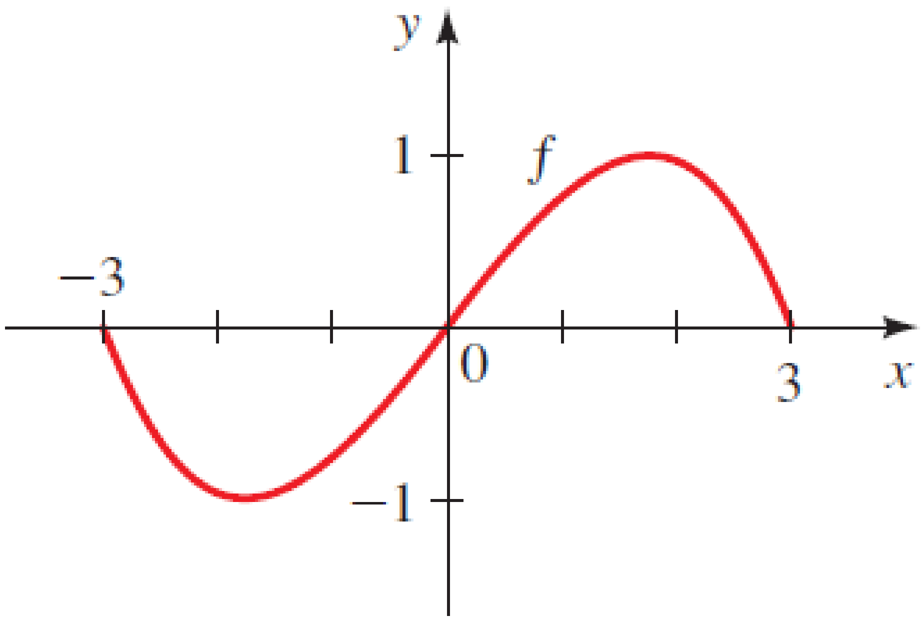
SOLUTION

1.4 Finding equations for transformations

SOLUTION

Graphing transformations

(a) $y = f(3x)$ **(b)** $y = f\left(\frac{1}{3}x\right)$

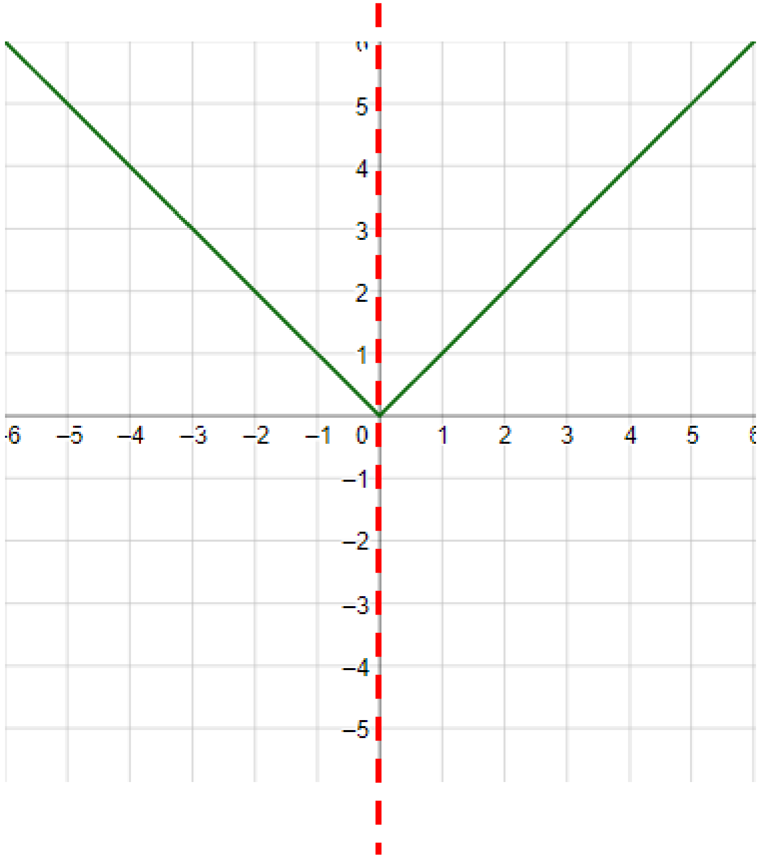


1.5 Graphing transformations

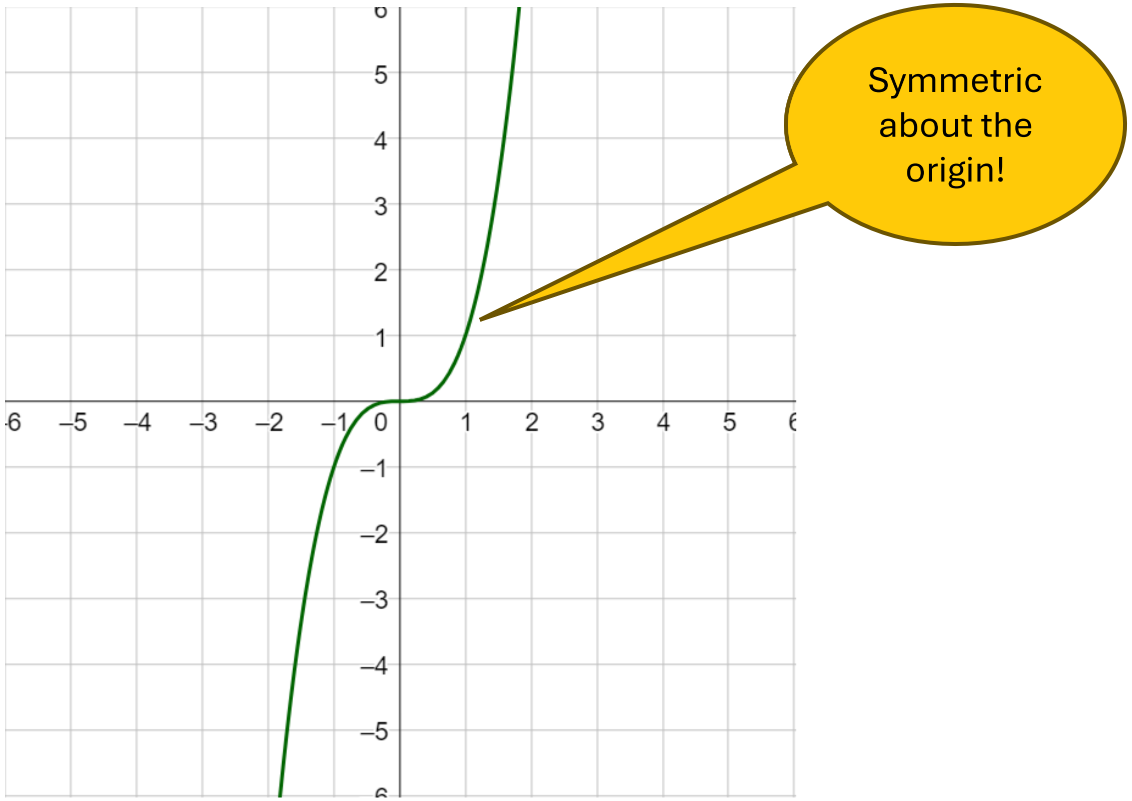
SOLUTION

Even and Odd functions

Graphs that are symmetric when reflecting over the *y*-axis are called **even functions**.



Functions that are symmetric after reflecting in both axes, or rotating 180°, are called **odd functions**.



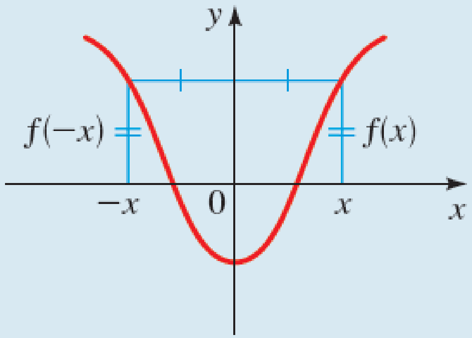
Even and Odd functions: Formal Definition

EVEN AND ODD FUNCTIONS

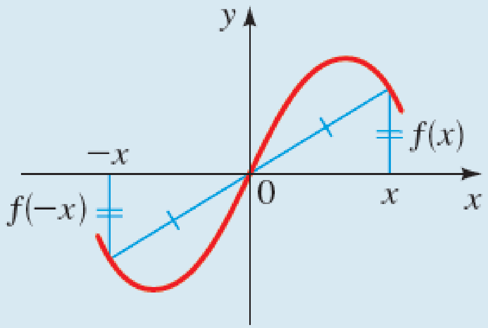
Let f be a function.

f is **even** if $f(-x) = f(x)$ for all x in the domain of f.

f is **odd** if $f(-x) = -f(x)$ for all x in the domain of f.



The graph of an even function is symmetric with respect to the y-axis.



The graph of an odd function is symmetric with respect to the origin.

Even and Odd functions

Determine whether the functions are even, odd, or neither even nor odd.

(a) $f(x) = x^5 + x$

(b) $g(x) = 1 - x^4$

(c) $h(x) = 2x - x^2$

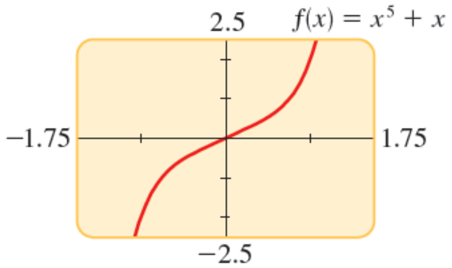
EXAMPLE

Even and Odd functions

$f(x) = x^5 + x$

$$f(-x) = (-x)^5 + (-x)$$
$$= -x^5 - x = -(x^5 + x)$$
$$= -f(x)$$

Therefore f is an odd function.

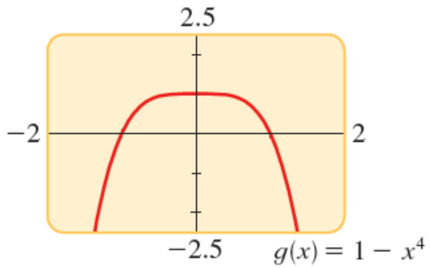


graph of f is **symmetric about the origin**

$g(x) = 1 - x^4$

$$g(-x) = 1 - (-x)^4 = 1 - x^4 = g(x)$$

So g is even.



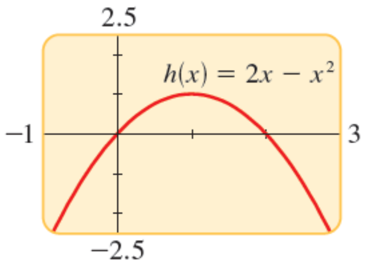
graph of g is **symmetric about the y-axis**

$h(x) = 2x - x^2$

$$h(-x) = 2(-x) - (-x)^2 = -2x - x^2$$

Since $h(-x) \neq h(x)$ and $h(-x) \neq -h(x)$,

Neither odd nor even



not symmetric about either the y-axis or the origin

EXAMPLE

Even and Odd functions

(a) $f(x) = x^4 - 4x^2$

(b) $f(x) = 3x^3 + 2x^2 + 1$

ACTIVITY 1.6

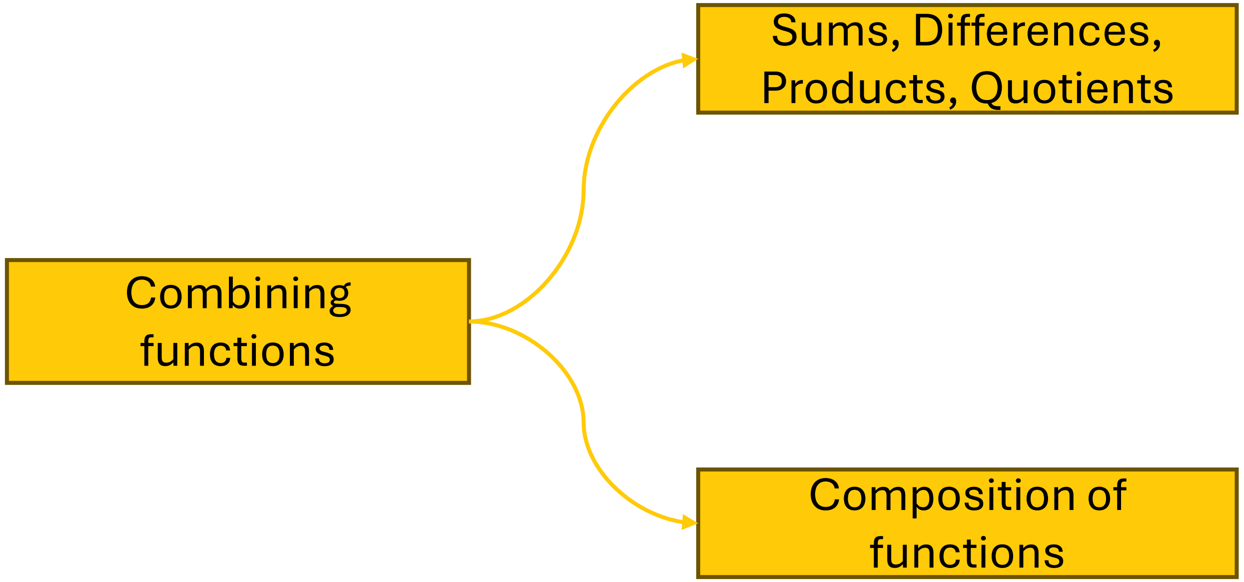
1.6 Even and Odd functions

SOLUTION

2. Combining functions

What are the different ways to combine functions to make new functions?

Combining Functions: Overview



Sums, Differences, Products and Quotients

ALGEBRA OF FUNCTIONS

Let f and g be functions with domains A and B. Then the functions $f + g$, $f - g$, fg, and f/g are defined as follows.

$(f + g)(x) = f(x) + g(x)$	Domain $A \cap B$
$(f - g)(x) = f(x) - g(x)$	Domain $A \cap B$
$(fg)(x) = f(x)g(x)$	Domain $A \cap B$
$\left(\dfrac{f}{g}\right)(x) = \dfrac{f(x)}{g(x)}$	Domain $\{x \in A \cap B \mid g(x) \neq 0\}$

Just like we can add, subtract, multiply and divide real numbers, we can **combine functions to create new functions**!

Example: Combination of functions and their domains

If $f(x) = x^2 - 1$ and $g(x) = x + 1$,

find

(a) $f + g$
(b) $f - g$
(c) $f \cdot g$
(d) $\frac{f}{g}$

EXAMPLE

Solution: Combination of functions and their domains

$f(x) = x^2 - 1$ and $g(x) = x + 1$

a. **Add the functions.**

$$f + g = (x^2 - 1) + (x + 1)$$
$$= x^2 + x$$

b. **Subtract the functions.**

$$f - g = (x^2 - 1) - (x + 1)$$
$$= x^2 - x - 2$$

c. **Multiply the functions.**

$$f \cdot g = (x^2 - 1)(x + 1)$$
$$= x^3 + x^2 - x - 1$$

d. **Divide the functions**.
(To simplify, factoring might be useful).

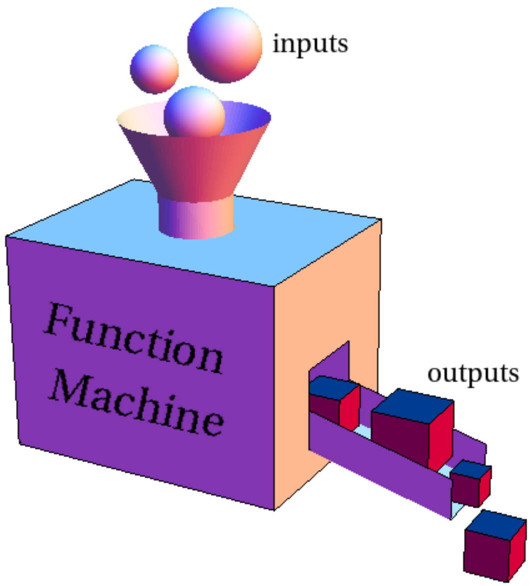
$$\frac{f}{g} = \frac{x^2 - 1}{x + 1}$$
$$= \frac{(x-1)(x+1)}{x+1}$$
$$= x - 1$$

only when $x \neq -1$

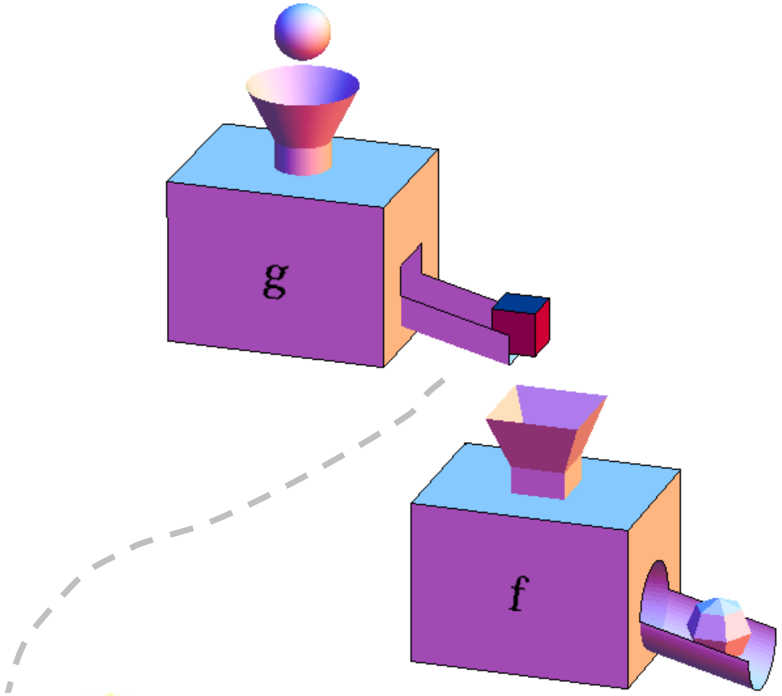
EXAMPLE

Composition of functions

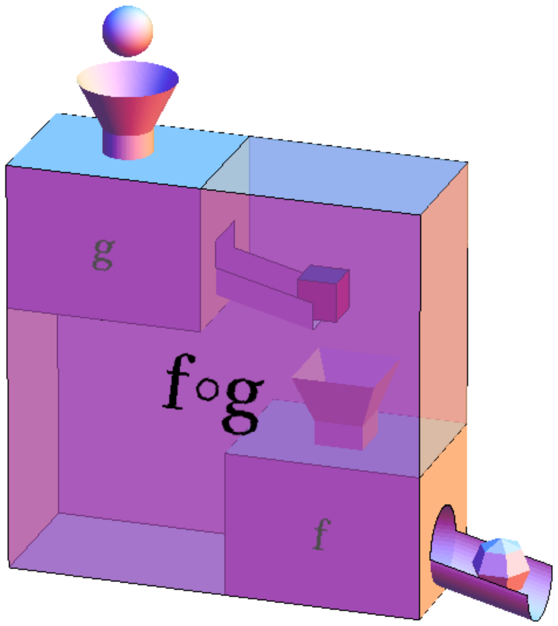
New function created by a composition is called a **composite function**



$f(g(x))$



$(f \circ g)(x)$



Whatever comes out of the output chute of g had better be able to fit into the input funnel of f.

→ make sure that the x is chosen such that g(x) is in the domain of f

$$(f \circ g)(x) = f(g(x))$$

[Nykamp DQ](http://mathinsight.org/function_machine), “The function machine.” From *Math Insight*. http://mathinsight.org/function_machine

Composition of functions: Formal definition

COMPOSITION OF FUNCTIONS

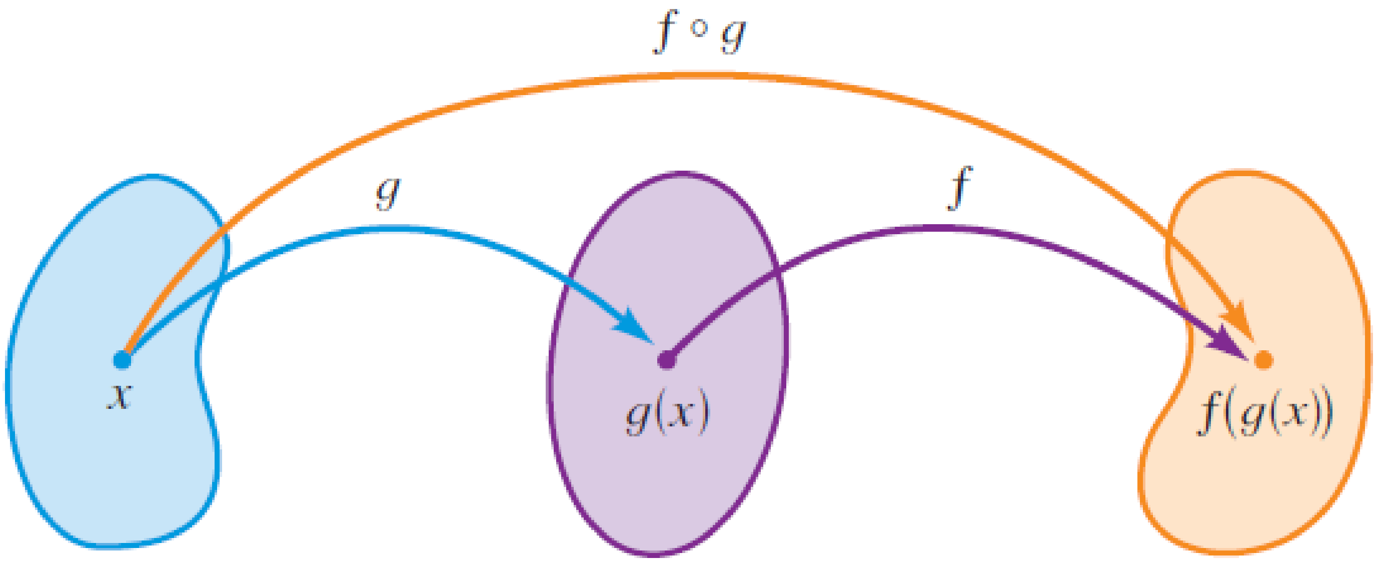
Given two functions f and g, the **composite function** $f \circ g$ (also called the **composition** of f and g) is defined by

$$(f \circ g)(x) = f(g(x))$$

Composition of functions: Domain of $(f \circ g)$

The domain of $(f \circ g)$ is the set of all x in the domain of g such that g(x) is in the domain of f.

That means, $(f \circ g)(x)$ is defined whenever both **g(x)** and **f(g(x))** are defined.



Combining functions

Find $f + g$, $f - g$, fg, and f/g and their domains.

(a) $f(x) = 3 - x^2, \quad g(x) = x^2 - 4$

(b) $f(x) = x, \quad g(x) = \sqrt{x}$

ACTIVITY 2.1

2.1a Combining functions

SOLUTION

2.1a (contd) Combining functions

SOLUTION

2.1b Combining functions

SOLUTION

2.1b (contd) Combining functions

SOLUTION

Composition of functions

Find the functions $f \circ g$, $g \circ f$, $f \circ f$, and $g \circ g$ and their domains.

(a) $f(x) = 6x - 5, \quad g(x) = \dfrac{x}{2}$

(b) $f(x) = \dfrac{1}{x}, \quad g(x) = 2x + 4$

ACTIVITY 2.2

2.2 Composition of functions

SOLUTION

SOLUTION

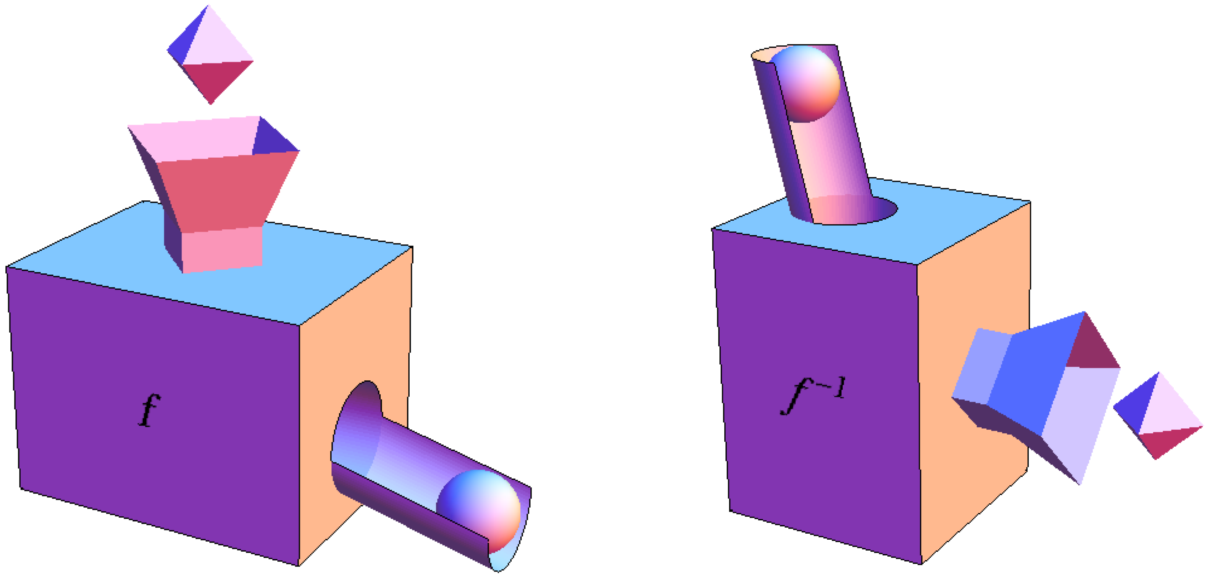
3. Inverse functions

Combining Functions: Overview

- Inverse of a function
  - Understanding inverse functions
    - Domain and Range
    - One-to-One functions & Horizontal Line test
  - Finding inverse of a function
    - For specific values,
    - From a table or graph
    - Algebraic method
  - Inverse function property
  - Graphing inverse functions

Inverse of a function

The *inverse* of a function is a **rule** that **acts on the output** of the function and **produces the corresponding input**.



“f inverse”

Or

“*Inverse of* f”

$$\boxed{f^{-1}}$$

A **backwards machine**
that
UNDOES
whatever processing f's function machine did

A bit like a vending machine that takes in the drink and dispenses money

Nykamp DQ, “The function machine inverse.” From *Math Insight*. http://mathinsight.org/function_machine_inverse

What inverse function is not!

if a function = f(x)

the inverse function is NOT $\frac{1}{f(x)}$

the inverse function is determined algebraically

What does the 'undo' part mean?

the inverse function must undo all of the **operations** from the original function

EXAMPLE

$f(x) = x + 2$

$f^{-1}(x) = x - 2$

If f^{-1} **adds** two to any input,

$f^{-1}(x)$must **subtract** two from any input

Finding inverse for specific values

If $f(1) = 5$, $f(3) = 7$, and $f(8) = -10$, find $f^{-1}(5)$, $f^{-1}(7)$, and $f^{-1}(-10)$.

SOLUTION From the definition of f^{-1} we have

$$\begin{aligned} f^{-1}(5) &= 1 \quad \text{because} \quad f(1) = 5 \\ f^{-1}(7) &= 3 \quad \text{because} \quad f(3) = 7 \\ f^{-1}(-10) &= 8 \quad \text{because} \quad f(8) = -10 \end{aligned}$$

$$f^{-1}(y) = x \iff f(x) = y$$

EXAMPLE

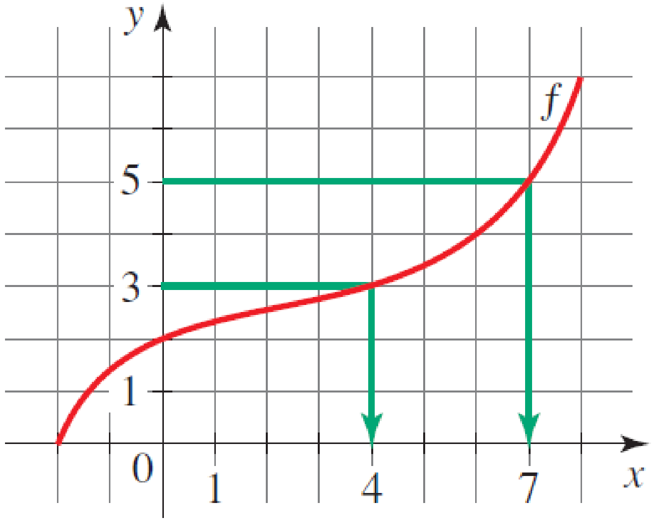
Finding values of an inverse function from table or graph

We can find specific values of an inverse function from a table or graph of the function itself.

(a) The table below gives values of a function h. From the table we see that $h^{-1}(8) = 3$, $h^{-1}(12) = 4$, and $h^{-1}(3) = 6$.

(b) A graph of a function f is shown in Figure 8. From the graph we see that $f^{-1}(5) = 7$ and $f^{-1}(3) = 4$.

x	$h(x)$
2	5
3	8
4	12
5	1
6	3
7	15

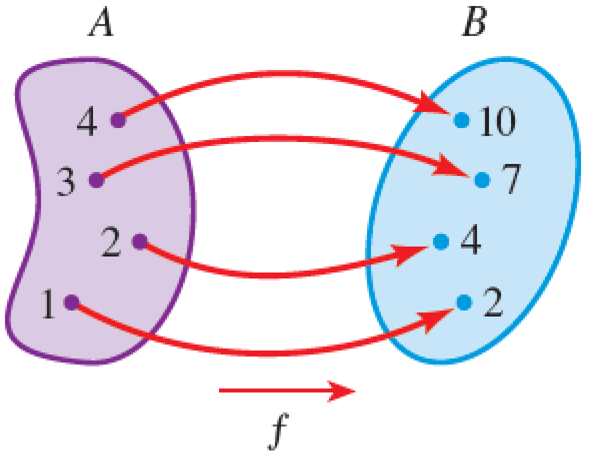


EXAMPLE

One-to-One Functions

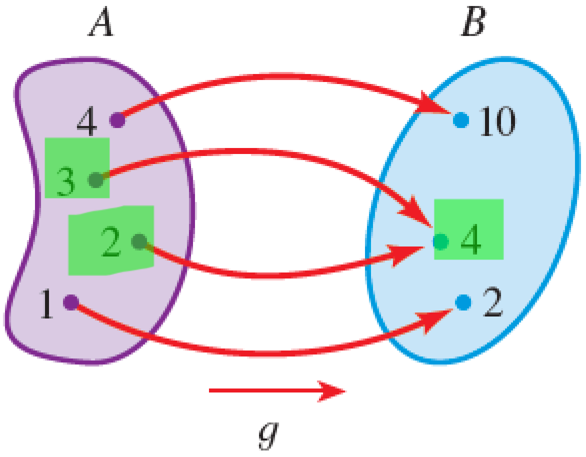
Not all functions have inverses. ***Only one-to-one functions have inverses.***

f never takes on the same value twice

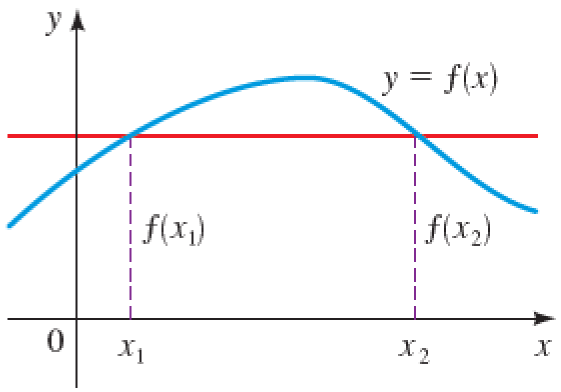


f is one-to-one

g does take on the same value twice



g is not one-to-one



Example of a function that gives the same output value for two different input
→ Not a one-to-one function!

DEFINITION OF A ONE-TO-ONE FUNCTION

A function with domain A is called a **one-to-one function** if no two elements of A have the same image, that is,

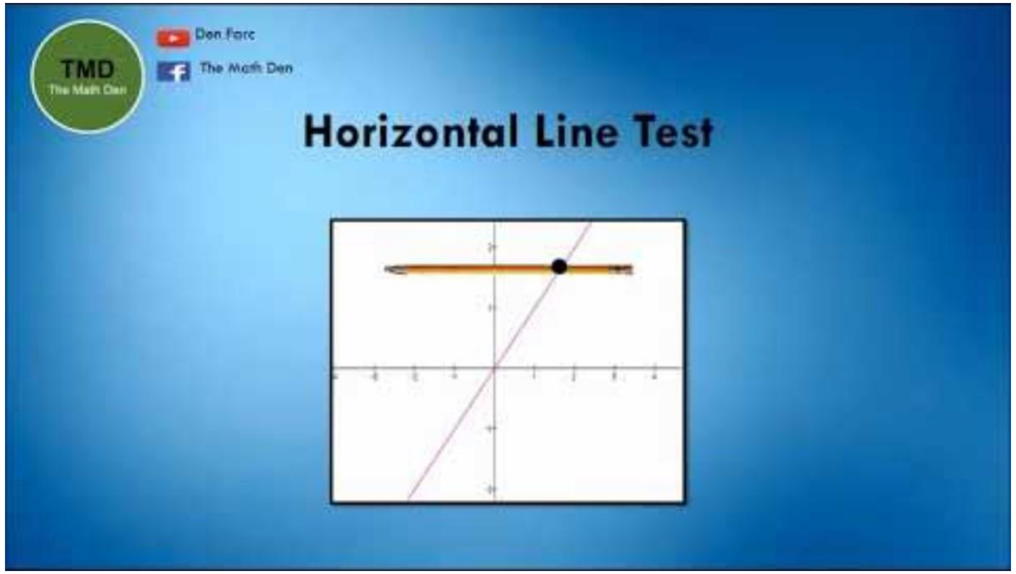
$$f(x_1) \neq f(x_2) \quad \text{whenever } x_1 \neq x_2$$

The Horizontal Line Test & one to one mapping

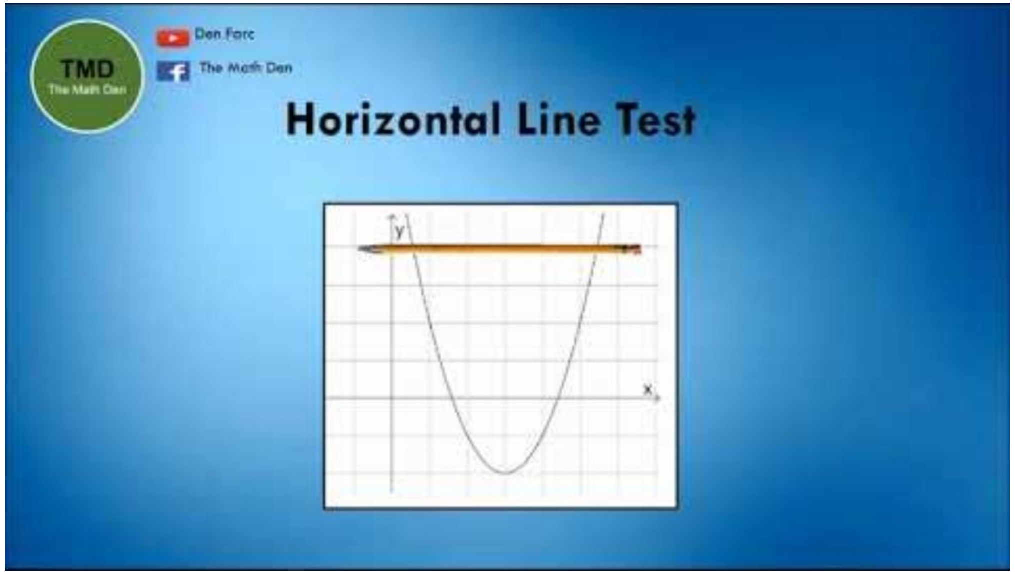
HORIZONTAL LINE TEST

A function is one-to-one if and only if no horizontal line intersects its graph more than once.

"If and only if"
"IFF"
"Bi-directional Conditional"



Example 1



Example 2

Inverse function property

INVERSE FUNCTION PROPERTY

Let f be a one-to-one function with domain A and range B. The inverse function f^{-1} satisfies the following cancellation properties:

$$f^{-1}(f(x)) = x \quad \text{for every } x \text{ in } A$$

$$f(f^{-1}(x)) = x \quad \text{for every } x \text{ in } B$$

Conversely, any function f^{-1} satisfying these equations is the inverse of f.

These properties indicate that f is the inverse function of f^{-1}, so we say that f and f^{-1} are *inverses of each other*.

You can use this property to **verify** whether your answer is correct, when computing inverses!

How to find the inverse of a function

Recipe

HOW TO FIND THE INVERSE OF A ONE-TO-ONE FUNCTION

1. Write $y = f(x)$.
2. Solve this equation for x in terms of y (if possible).
3. Interchange x and y. The resulting equation is $y = f^{-1}(x)$.

Rationale behind the recipe

$y = f(x)$ we know this from the definition of functions

Therefore, we get

$f^{-1}(y) = x$ using algebra

We want to determine **what is $f^{-1}(x)$**

So we just **swap** x and y to get the desired equation!

$f^{-1}(x) = y$

Finding the Inverse of a Function

Find the inverse of the function $f(x) = 3x - 2$.

SOLUTION First we write $y = f(x)$.

$$y = 3x - 2$$

Then we solve this equation for x:

$$3x = y + 2 \qquad \text{Add 2}$$

$$x = \frac{y + 2}{3} \qquad \text{Divide by 3}$$

Finally, we interchange x and y:

$$y = \frac{x + 2}{3}$$

Therefore, the inverse function is $f^{-1}(x) = \dfrac{x + 2}{3}$.

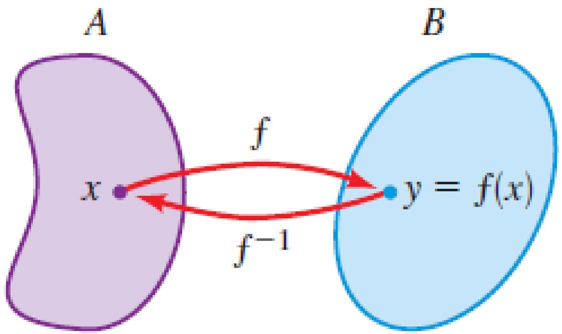
You can check your answer using the inverse function property!

EXAMPLE

Understanding inverse functions: Domain and Range

domain of $f(x)$ = range of $f^{-1}(x)$

domain of $f^{-1}(x)$ = range of $f(x)$



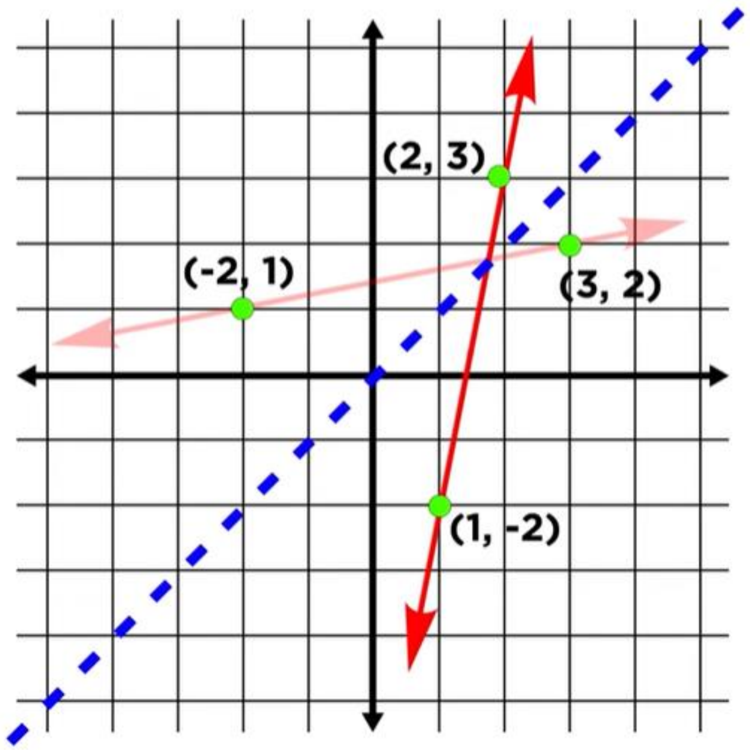
$$\boxed{\begin{array}{c} f(x) = (x, y) \\ f^{-1}(x) = (y, x) \end{array}}$$

f is the set of ordered pairs (x, y),

f^{-1} is the set of ordered pairs (y, x)

Graphs of inverse functions

$$f(x) = (x, y)$$
$$f^{-1}(x) = (y, x)$$

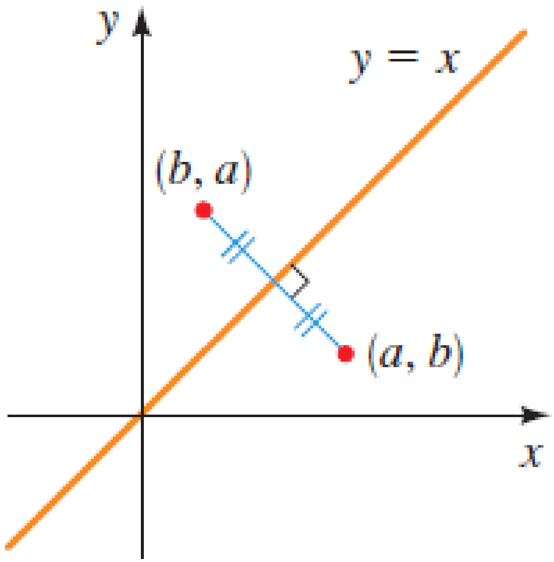


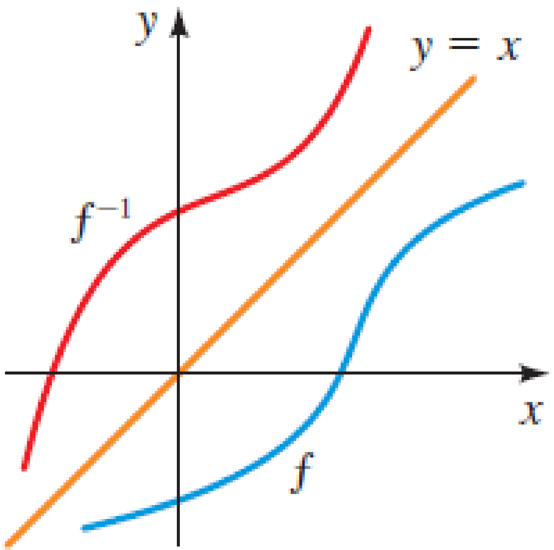
every point on a function will have its **coordinates reversed** on the inverse function

to graph the **inverse** of a function we simply **reflect** the function across the line $y = x$

Graphing the inverse of a function

The graph of f^{-1} is obtained by reflecting the graph of f in the line $y = x$.





Finding values of an inverse function

Assume that f is a one-to-one function.

(a) If $f(5) = 18$, find $f^{-1}(18)$.

(b) If $f^{-1}(4) = 2$, find $f(2)$.

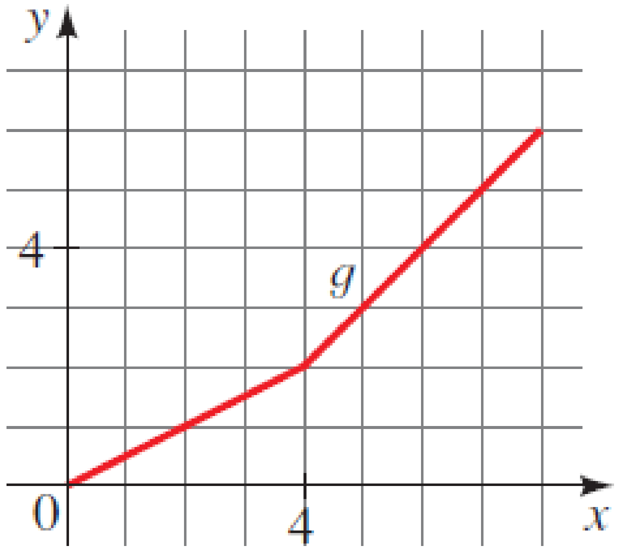
ACTIVITY 3.1

3.1 Finding values of an inverse function

SOLUTION

Finding values of an inverse from a graph

(a) $g^{-1}(2)$ **(b)** $g^{-1}(5)$ **(c)** $g^{-1}(6)$



ACTIVITY 3.2

3.2 Finding values of an inverse from a graph

SOLUTION

Inverse function property

Use the Inverse Function Property to show that f and g are inverses of each other

$$f(x) = x^3 + 1; \quad g(x) = (x - 1)^{1/3}$$

3.3 Inverse function property

SOLUTION

Find the inverse of a function

(a) $f(x) = 7 - 5x$

(b) $f(x) = \dfrac{x - 2}{x + 2}$

3.4 Find the inverse of a function

SOLUTION

3.4 Find the inverse of a function

SOLUTION

Key points we covered today

1 Functions **represent relationships** where each input (x-value) corresponds to **exactly** one output (y-value).

2 Graphing functions visually represents these relationships on a coordinate plane.

3 Functions can be transformed through translations (shifts), reflections, stretches, and compressions.

4 Functions can be combined using operations like addition, subtraction, multiplication, division, and composition.

5 An inverse function **undoes** the effect of another function.

- To find the inverse, **switch the roles of x and y** in the original function and solve for y.
- The graph of an inverse function is **symmetric** to the original function about the line $y = x$.